

Oct 23, 2025

Meeting Oct 23, 2025 at 15:25 CEST

Meeting records  Recording

Summary

Clément Lacaille introduced the concept of adding memory to agents using databases, specifically vector databases, to enable personalized advice, initially using a personal investment advisor example and later switching to a meal planning agent. Clément Lacaille guided Murphy Odion Usunobun, Naweed Zahed, Naim Burrack, Leon Winde, Sara Hofmann, and Chinedu Kirian Onwuzuruike through setting up a Pinecone account, creating an index for meal planning data, configuring its settings, and integrating it into an n8n workflow to upload Sven Lück's meal preference data. The workflow, including Pinecone and OpenAI embeddings, was successfully executed by Murphy Odion Usunobun and Naweed Zahed, verifying data upload to Pinecone.

Details

Notes Length: Standard

- **Introduction to Agent Memory** Clément Lacaille introduced the concept of adding memory to agents, highlighting that an agent typically has four components: an LLM (brain), a system prompt for instructions, APIs for external tools, and a database/memory. He explained that memory allows agents to store and retrieve data, unlike logical reasoning alone, and that the focus for the next two days would be on understanding vector databases.
- **Objective of Learning Databases** Clément Lacaille clarified that the main objective is to understand database concepts for agents, especially vector

databases, which are crucial for building agents. He explained that understanding this concept is essential for building any future agents.

- **Personal Investment Advisor Agent** Clément Lacaille elaborated on the goal of creating a personal investment advisor agent that can access and utilize an individual's historical investment data to provide personalized advice, unlike current agents that only use publicly available information. This aims to make the agent more intelligent by providing context from personal investment history.
- **Clarifying the Database Concept** Clément Lacaille asked for clarification from participants regarding the objective of adding a database. Sven Lück initially misunderstood, thinking it was for tracking token usage, but then understood it was for creating a personalized investment plan based on stored data. Clément Lacaille further clarified that the goal is to make the agent more intelligent by providing it with personal historical data, like past stock investments, to offer tailored recommendations.
- **Switching to Meal Planning Agent Example** Due to some complexity with the investment example, Clément Lacaille suggested switching the context to a meal planning agent, which they believed would be more relatable and easier to understand. The goal for this agent would be to provide personalized meal suggestions based on the user's past meal consumption and preferences.
- **Understanding Vector Databases and Embeddings** Clément Lacaille directed Murphy Odion Usunobun to ask the agent to explain what a vector database and embeddings are, assuming the user is a 10-year-old and providing food-related examples. He emphasized the importance of understanding the theoretical concepts of how agents retrieve and insert data from a database.
- **Demonstration of Meal Planning Agent with Vector Database** Clément Lacaille guided the agent to provide a demo of how a meal planning agent uses a vector database. The demo illustrated how the agent "remembers" previous meals by storing them as numerical lists (taste points, healthiness), searches for similar and healthy options based on new queries, and then provides personalized recommendations.
- **Analogy for Personalized Recommendations** To further clarify the concept, Clément Lacaille used the analogy of Spotify or Netflix for meals, explaining that the agent adapts recommendations based on past consumption data. He confirmed that the goal is to create a personalized experience by taking the user's past data into account.

- **Creating Fake Meal Data for Practice** Clément Lacaille instructed the group to create a fake document with detailed meal consumption data for the past three months, including ingredients, cooking time, and spice levels, to simulate a real database for the exercise. This fake data will be used to demonstrate how the agent would personalize meal suggestions.
- **CSV Document Generation** Clément Lacaille guided Murphy Odion Usunobun to generate a CSV document with "three meals a day" using a publicity tool. However, the tool did not provide a direct link to the document, prompting Clément Lacaille to suggest requesting the data directly due to perceived limitations of the free version. Ultimately, they decided to use a pre-existing document provided by Sven Lück from Slack as a temporary solution.
- **Pinecone Account Setup** Clément Lacaille instructed Murphy Odion Usunobun and other participants to set up a Pinecone account. This involved logging in with Google, selecting "small personal project" as the purpose of use, and choosing "no coding" for preferred coding language. Participants were also guided to fill out survey questions, such as "agents" for use cases and "learn about vector search" for their goal.
- **API Key Retrieval** After setting up the Pinecone account, Clément Lacaille directed participants to copy their API key. Murphy Odion Usunobun was instructed to save the API key in a document named "pine cone" for future reference. Clément Lacaille emphasized that all participants should complete this step, including logging in with Google and copying the API key, before proceeding.
- **Creating a Pinecone Index** Clément Lacaille guided Murphy Odion Usunobun to create a new index in Pinecone, naming it "meal planning data". They stressed the importance of following instructions carefully to avoid errors that would necessitate recreating the database. Naim Burrack and Leon Winde also provided assistance regarding the naming conventions and character limitations for the index.
- **Index Configuration** When configuring the Pinecone index, Clément Lacaille specified important settings. Participants were instructed to select "Text embedding three small text" as the model and set the dimension to "1536". Naim Burrack highlighted the need to remove spaces and convert uppercase letters to lowercase in the index name.

- Understanding Pinecone's Use Cases** Clément Lacaille explained the practical applications of Pinecone, highlighting its utility in professional settings. They described how Pinecone can store unstructured data, such as descriptions and image links for character models, allowing agents to retrieve relevant context for tasks like image generation. This demonstrated Pinecone's role in retrieving context from large, unorganized data sets.
- Creating a New Workflow** To automate the process of inserting data into Pinecone, Clément Lacaille initiated the creation of a new workflow in n8n. The workflow was named "Murphy_ Pine Cone database," and its purpose was to insert data into the newly created Pinecone index. The first step in the workflow was set to "Trigger manually".
- Uploading Sven's File** Clément Lacaille instructed Murphy Odion Usunobun to download Sven Lück's meal preference data file from Slack. Murphy Odion Usunobun then uploaded this PDF file to their Google Drive, which was linked to their n8n account. Sara Hofmann confirmed the file was in PDF format, and Clément Lacaille noted that working with PDF files was easier for this process.
- Downloading the File in the Workflow** Within the workflow, Murphy Odion Usunobun was guided to add a Google Drive node to "Download a file". They selected "From list" and searched for the "Meal plan fake data" file in their Google Drive. Naim Burrack assisted in ensuring Murphy Odion Usunobun was in the correct Google Drive before executing the step.
- Connecting Pinecone to the Workflow** Clément Lacaille guided Murphy Odion Usunobun to add a "Pinecone Vector Store" node to the workflow, specifically choosing the "Add documents to vector store" action. This step was crucial for connecting the workflow to the Pinecone database. Participants were instructed to connect their Pinecone account by creating a new credential and inputting their API token.
- Configuring Pinecone Credential** When setting up the Pinecone credential, Clément Lacaille advised participants to name it appropriately, such as "Murphy's Pine API" or "N's Pine Cone API". Murphy Odion Usunobun demonstrated saving the credential with a personalized name. They then selected the "meal planning data" index from the list, linking the workflow to the correct database.
- Setting up Embeddings and Document Loader** Clément Lacaille instructed participants to add an "Open AI Embeddings" node to the workflow, ensuring the model chosen was "text embedding three small". Subsequently, a "Default data

loader" node was added, configured to use "Binary" data and "Custom" text splitting. Naim Burrack clarified that they had followed the instructions correctly despite initial confusion.

- **Workflow Execution and Data Verification** After all the setup steps were completed, Clément Lacaille instructed Murphy Odion Usunobun to execute the workflow. The workflow ran successfully, and Murphy Odion Usunobun confirmed that the data had been uploaded to their Pinecone account by checking the Pinecone website. Clément Lacaille assigned homework, encouraging participants to explore the uploaded data and research concepts like vector databases, vectoring, and embedding.
- **Troubleshooting and Assistance** Clément Lacaille provided individual assistance to participants experiencing issues during the setup. Sara Hofmann had an authentication issue with Google Drive, which Clément Lacaille resolved by guiding them to modify the client ID and re-authenticate. Chinedu Kirian Onwuzuruike, whose system froze and deleted everything, received step-by-step guidance to rebuild their workflow from scratch. Naweed Zahed also received assistance in locating their Google Drive credentials and correctly configuring their Pinecone API key.
- **Setting up Pinecone Database** Clément Lacaille guided Naweed Zahed through setting up a Pinecone database, emphasizing the need to create a database record before entering data. Naweed Zahed was instructed to create an index with specific configurations and dimensions, including selecting "1536" for the dimension setting.
- **Troubleshooting and Connection** Clément Lacaille assisted Naweed Zahed in resolving connection issues by retrying actions and reconfiguring the Pinecone vector store. They successfully connected the Pinecone account and integrated the documents into the vector store by selecting the correct credential.
- **Workflow Execution** Clément Lacaille directed Naweed Zahed to draw lines connecting the embeddings to the OpenAI in the workflow. After completing the connections, Naweed Zahed executed the workflow, and it was successfully completed.

Suggested next steps

- ☐ Clément Lacaille will switch the focus of the current exercise from building an investment agent to building a meal planning agent, as it is considered more relatable and simpler to understand for everyone.
- ☐ Clément Lacaille will share some links and a sample document tomorrow to make the example clearer.
- ☐ The group will try to go and look at the uploaded Pine Cone data, read it, and research vector data, vectoring, and embedding.
- ☐ Clément Lacaille will follow up on the class tomorrow.
- ☐ The group will fill out the Google form for last week if they haven't yet.

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